

Parsa Rangriz

CONTACT INFORMATION	Department of Mathematics, UC San Diego, 9500 Gilman Drive, La Jolla, CA, USA, 92093 Office: HSS 4047	prangriz@ucsd.edu https://prangriz.github.io
EDUCATION	University of California, San Diego Ph.D. in Mathematics University of Waterloo Master of Mathematics in Statistics <i>Advisor:</i> Aukosh Jagannath Sharif University of Technology Bachelor of Science in Physics <i>Extra:</i> Minor in Mathematics	2025 - Present 2023 - 2025 2018 - 2023
PUBLICATIONS	Parsa Rangriz, <i>Limit theorems for stochastic gradient descent in high-dimensional single-layer networks</i> , Electronic Communications in Probability, 31, No. 15 (2026). Parsa Rangriz, <i>High-dimensional scaling limits of online stochastic gradient descent in single-index models</i> , Master of Mathematics Thesis, University of Waterloo (2025).	
EMPLOYMENT	Teaching Assistant , UC San Diego MATH 18 (<i>Fall '26</i>); PHYS 1CL (<i>Summer '26</i>); MATH 193A (<i>Summer '26</i>); MATH 180A (<i>Spring '26</i>); MATH 10B (<i>Winter '26</i>); MATH 11 (<i>Fall '25</i>) Graduate Research Student , University of Waterloo <i>Supervisor:</i> Aukosh Jagannath Worked on high-dimensional probability, scaling limits, random matrices, and mathematical foundations of machine learning. Teaching Assistant , University of Waterloo STAT 333 (<i>Winter '25, Winter '24</i>); STAT 433 (<i>Fall '24</i>); STAT 330 (<i>Fall '23, Spring '24</i>); STAT 231 (<i>Spring '24</i>); STAT 230 (<i>Spring '24</i>); STAT 202 (<i>Fall '23</i>) Summer Research Intern , EPFL <i>Supervisor:</i> Lenka Zdeborova Worked on statistical physics methods for machine learning and combinatorial optimization, e.g., mean-field theory, replica method, and message-passing algorithms.	2025 - Present 2023 - 2025 2023 - 2025 2022
AWARDS & HONORS	First-Year Departmental Fellowship . UC San Diego Graduate Research Studentship . University of Waterloo International Master's Award of Excellence . University of Waterloo Master of Math Entrance Scholarship . University of Waterloo Summer@EPFL Fellowship . EPFL Silver Medal in the Iranian National Physics Olympiad	2025 - 2026 2023 - 2025 2023 2023 2022 2017